

Fabric Security

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Agenda

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- Network Security
- Tenant, Workspace Private Links
- Entra Conditional Access
- Authentication & Authorization
- Workspace Roles
- Item Level Access

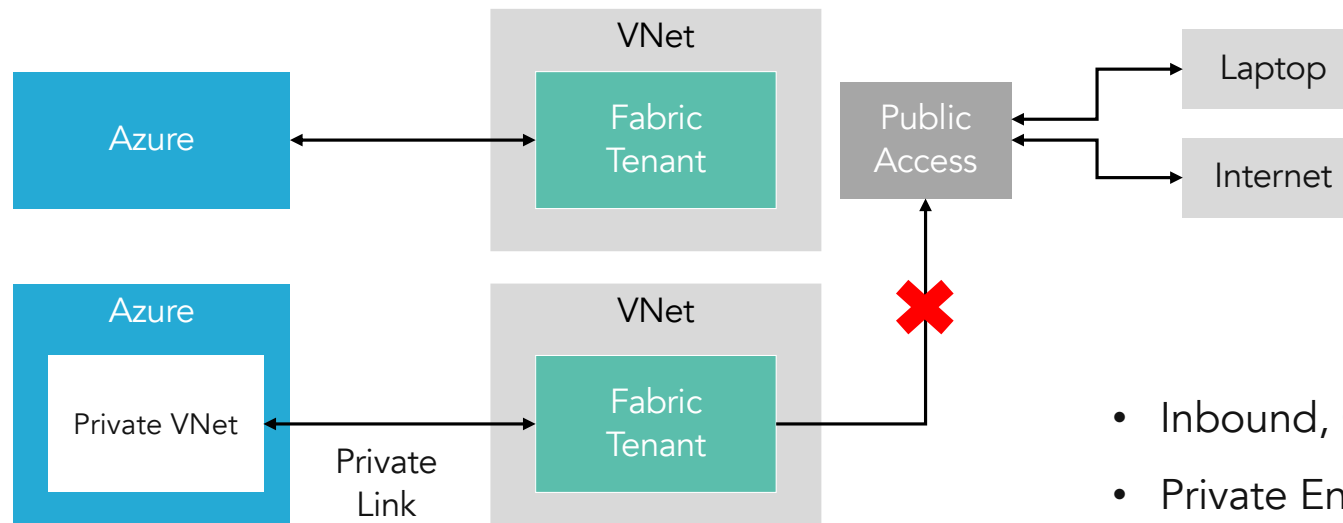
Managed Security

- Fabric is a SaaS Data Platform.
- Following are managed automatically:
 - OS Patches.
 - Software Upgrades.
 - Software patches.

Fabric Tenant Settings

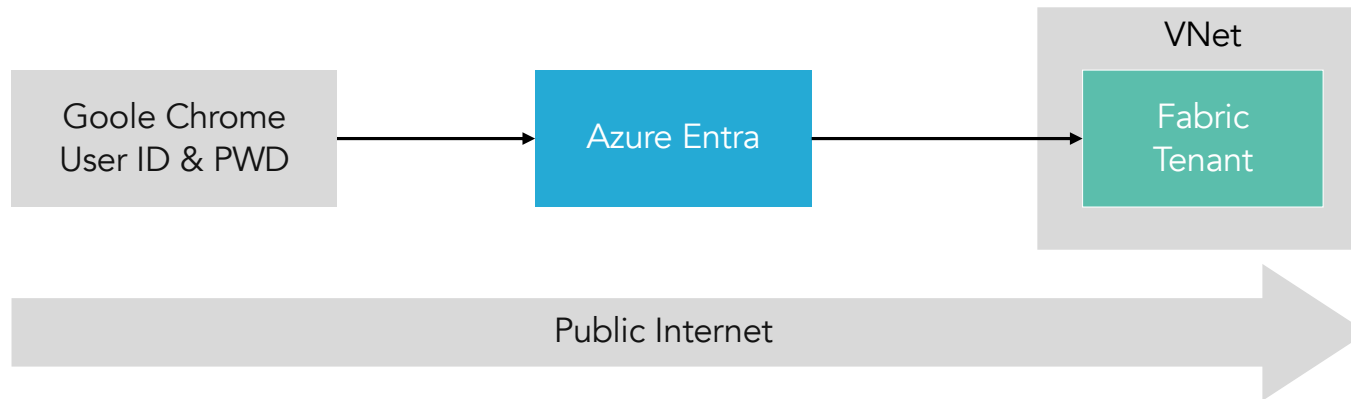
- Fabric Item settings.
- Workspace settings.
- Sensitivity settings.
- Data sharing settings.
- Integration with 3rd party tools, APIs.
- Developer and Admin API settings – Service Principal.
- Networking settings.
- OneLake & Git Integration settings.
- Azure connections.
- Workspace Access (Authentication & Authorization).
- Fabric Identities (Authentication & Authorization).

Network (Infra) Security



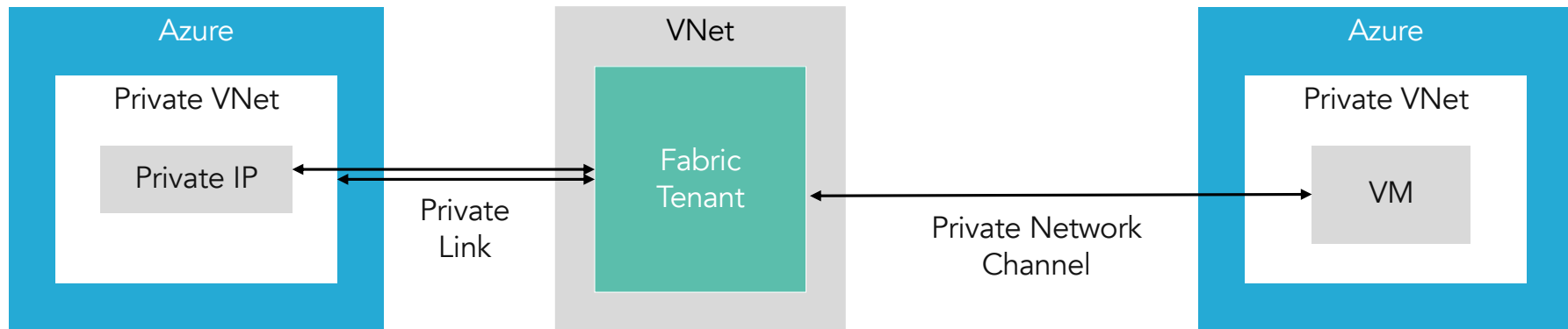
- Inbound, Outbound network security.
- Private Endpoints, Private Links.
- Data Gateway, VNet Gateway.

Fabric Access Flow



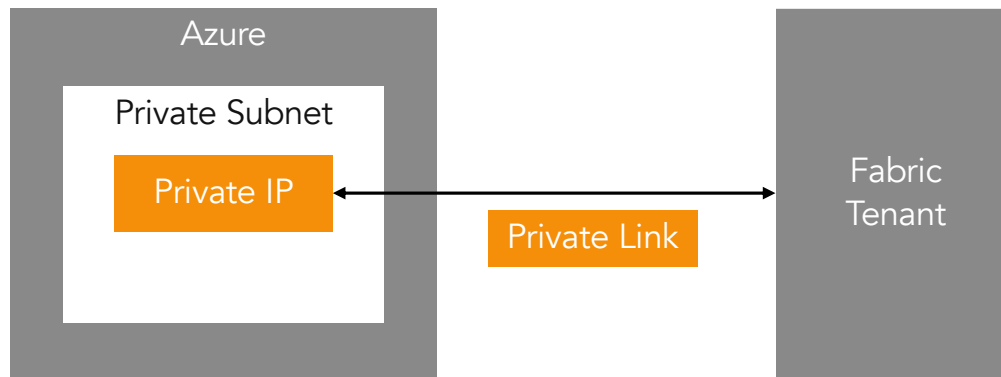
- Open Laptop browser.
- Provide user id (3aayaam-fabric-admin) and pwd.
- Authenticated by Entra.
- Access is provided to Fabric console from Internet.

Fabric Private Endpoint



- Private Endpoint is created for Fabric.
- Private IP is in the Private VNet. It is used to access Fabric via Azure backbone network.
- Access from Internet is removed.
- Two types of Private Link – Tenant Level, Workspace Level.

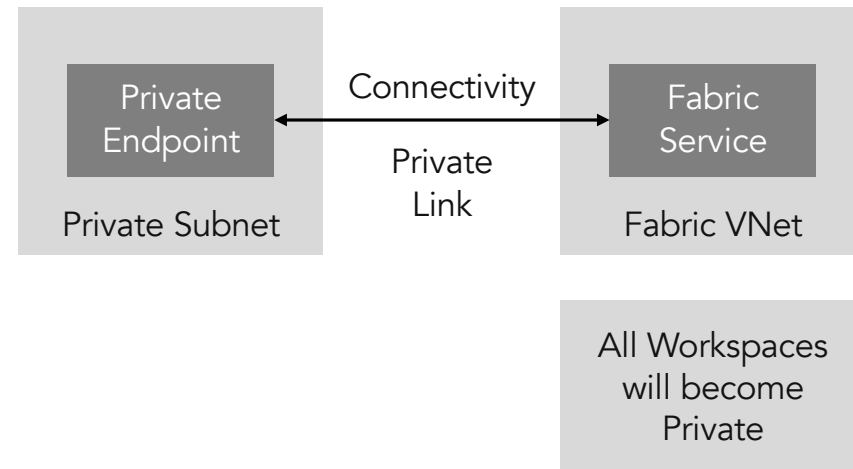
Private Endpoint, Private Link



- Private Link (Private Link Service) is the private connection between Fabric and Azure Subnet using Azure Network Banckbone.
- Private Endpoint is the Private IP that is used to access Fabric over private connection.

Tenant Level Private Link

- Private Link and Private Endpoint.
- Set up using Tenant settings.
- Enable Tenant-level Private Link.
- Block Public Internet Access.
- Each Workspace gets their own VNet.
- Private Link supported items in Fabric:
 - OneLake.
 - SQL Analytics Endpoint – Lakehouse.
 - SQL Analytics Endpoint – Warehouse.
 - Fabric Notebook, Spark Job Definition. Starter pools cannot be used.
 - Dataflow Gen2, Pipeline.
 - Eventstream & Eventhouse.
 - Mirrored database.



Steps : Tenant Private Link

- 1) Enable Tenant-Level Private Link.
- 2) Azure Portal - Create Private Link Service for Fabric.
- 3) Create VNet.
- 4) Create a Private Subnet.
- 5) Create Private Endpoint.
 - i. With Resource Type Microsoft.PowerBI/privateLinkServicesForPowerBI.
 - ii. Integration with Private DNS.
- 6) Create VM.
 - i. Check IP addresses of Fabric services.
 - ii. `app.fabric.Microsoft.com` to access Fabric portal.
- 7) Access Fabric from VM.

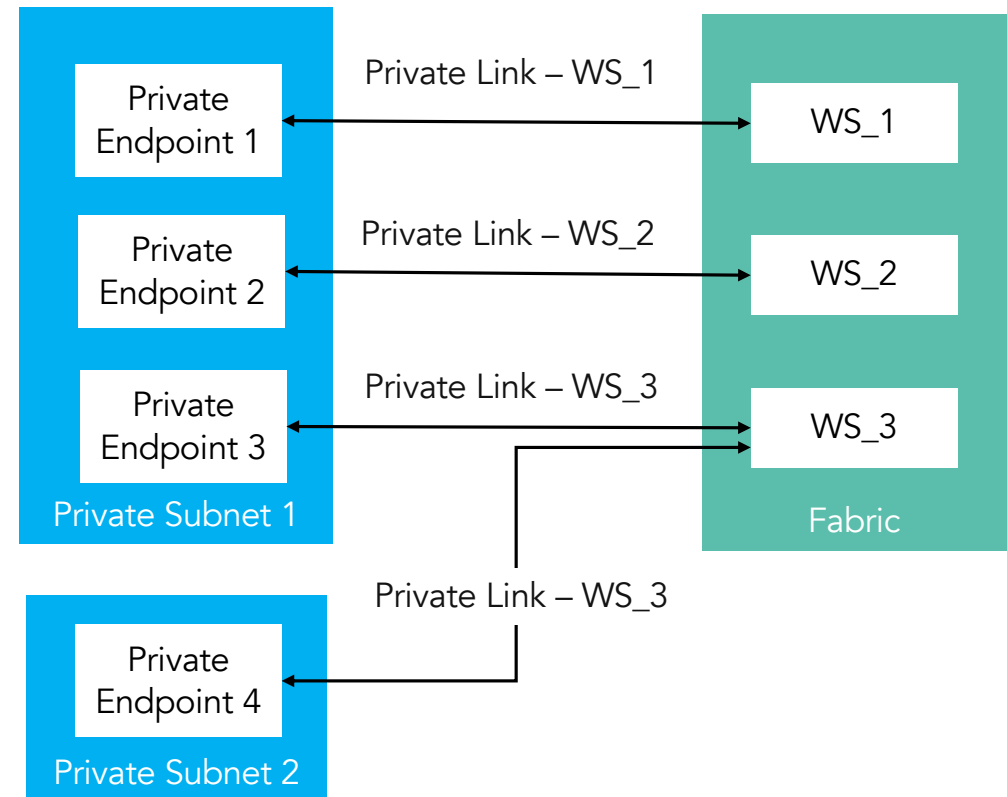


Tenant PLS Template

```
{
  "$schema": "http://schema.management.azure.com/schemas/xxxx",
  "contentVersion": "1.0.0.0",
  "parameters": {},
  "resources": [
    {
      "type": "Microsoft.PowerBI/privateLinkServicesForPowerBI",
      "apiVersion": "2020-06-01",
      "name": "<resource-name>",
      "location": "global",
      "properties": {
        "tenantId": "<tenant-object-id>"
      }
    }
  ]
}
```

Workspace Level Private Link

- Private Link and Private Endpoint is at Workspace level.
- Each Workspace should have its own Private Link and Private Endpoint.
- There is one-to-one mapping between Workspace and Private Link Service.
- Same Private Link can be used by multiple PEPs from different VNets.



Steps : Workspace Private Link

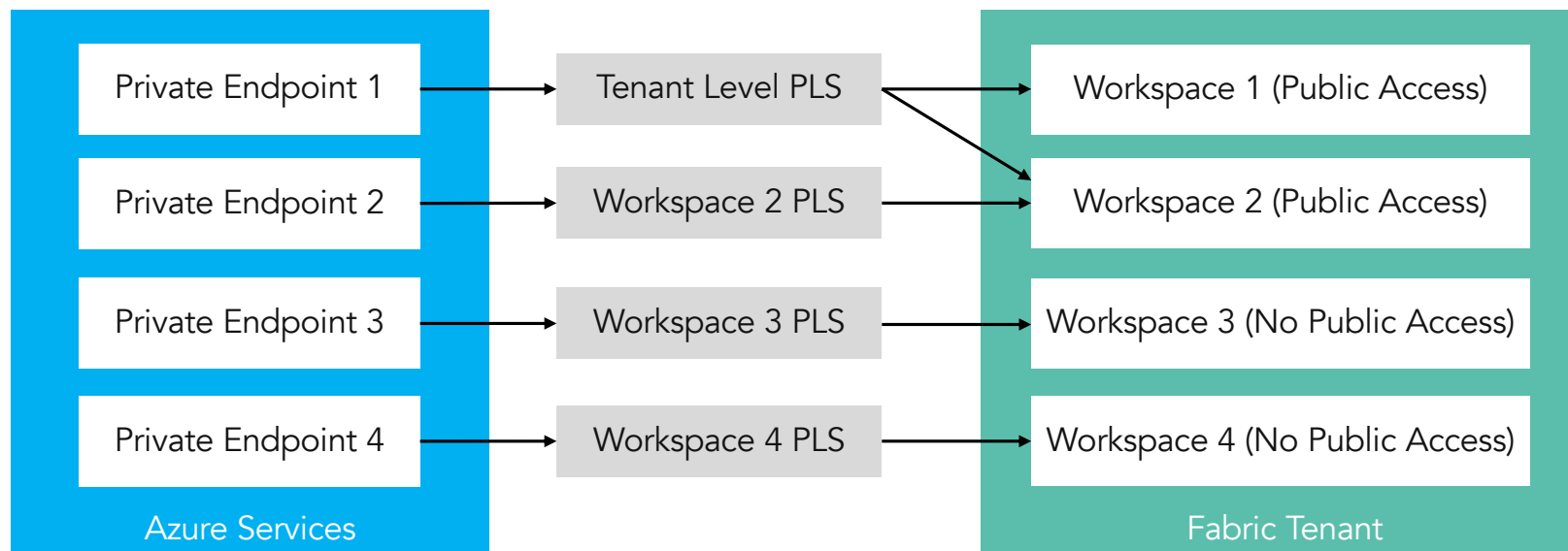
- 1) Workspace should have Fabric Capacity. Trial cannot be used.
- 2) Enable Workspace-level inbound network rule at Tenant settings..
- 3) Enable Workspace-level outbound network rule at Tenant settings.
- 4) Block outbound public access at Workspace settings.
- 5) Enable allow access only from workspace level private links.
- 6) Create Private Link Service.
- 7) Use VNet, private subnet. Create private endpoint.
- 8) Check private access - nslookup {workspaceid}.z{xy}.w.api.fabric.microsoft.com
- 9) Open workspace.

Workspace PLS Template

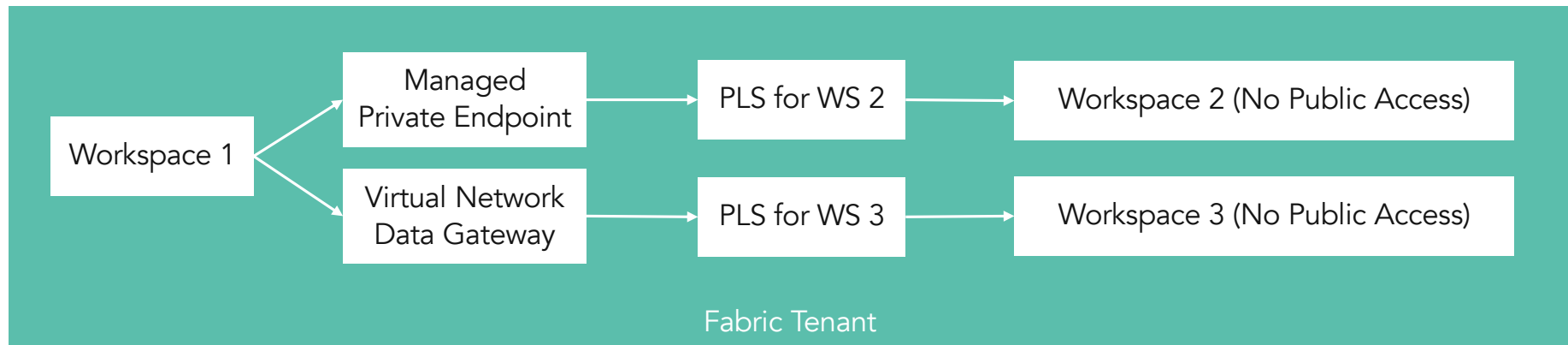
```
{
  "$schema": "http://schema.management.azure.com/schemas/2015-01-",
  "contentVersion": "1.0.0.0",
  "parameters": {},
  "resources": [
    {
      "type": "Microsoft.Fabric/privateLinkServicesForFabric",
      "apiVersion": "2024-06-01",
      "name": "<resource-name>",
      "location": "global",
      "properties": {
        "tenantId": "<tenant-id>",
        "workspaceId": "<workspace-id>"
      }
    }
  ]
}
```

Workspace + Tenant PL

Access to Fabric	Tenant Level Private Link with Internet Allowed	Tenant Level Private Link with Internet Blocked
Workspace Level Private Link with "Allow All" Enabled	Allowed	Allowed using TLPL
Workspace Level Private Link with "Allow All" Disabled	Allowed using WLPL	Allowed using WLPL



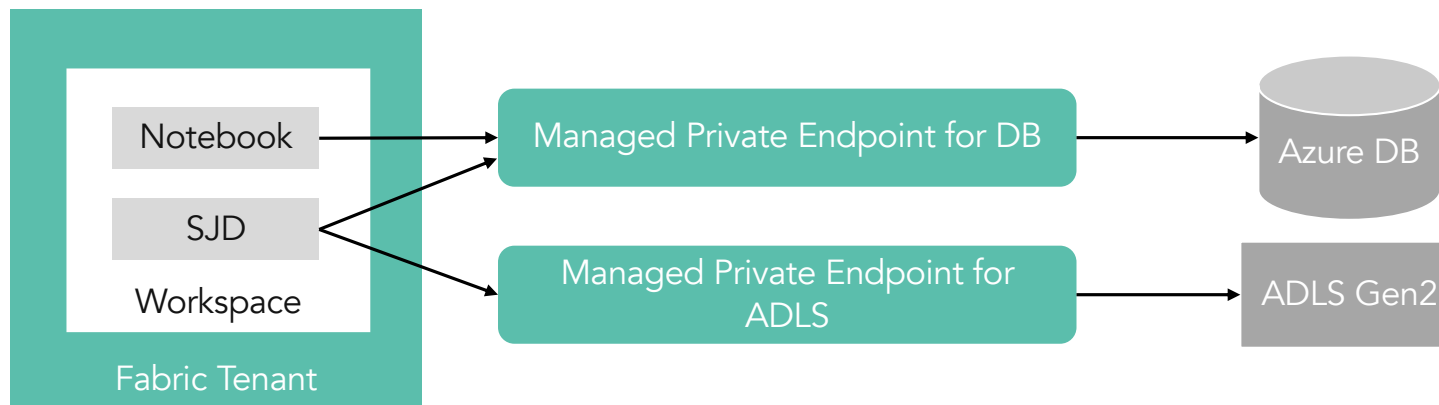
Cross Workspace Access



- Managed Private Endpoint along with Workspace-level Private Link Service.
- VNet Data Gateway along with Workspace-level Private Link Service.
- Managed Private Endpoint creates Managed VNet in Fabric for the workspace. This VNet can be used to run spark clusters in isolation.

Managed VNet, Private Endpoint

- Used to create private access to data sources.
- Created when Managed Private Endpoint is created.
- Created when Private Link is created for Tenant or Workspace.
- Used to control outbound access from workspaces. Block public/internet access.



Entra Conditional Access

- Can be used to control access to Fabric on conditions.
- Specific users and groups can be added.
- Fabric URLs can be added in the ALLOWLIST.
 - Platform.
 - OneLake.
 - Pipeline.
 - Lakehouse and Notebook.
 - Warehouse.
 - Eventstream & KQL Database.
- Azure Service tags can also be used to control access.

Resource/Item Level Security

Authentication & Authorization

- Entra ID Authentication.
- User had to be added in Entra.
- Authorization:
 - Workspace Roles.
 - Service Principal.
 - Workspace Identity.

Workspace Roles

- Viewer - Can **view** all content in the workspace, but can't modify it.
- Contributor - Can **view** and **modify** all content in the workspace.
- Member - Can **view**, **modify**, and **share** all content in the workspace.
- Admin - Can **view**, **modify**, **share**, and **manage** all content in the workspace, including **managing permissions**.
- Hands-On:
 - Create a user (fabric-dev-user) in Entra.
 - Provide Viewer access to fabric-dev-user.
 - Modify to Contributor access to fabric-dev-user.
 - Modify to Member access to fabric-dev-user.

Workspace Roles

Capability	Admin	Member	Contributor	Viewer
Delete the workspace	✓	✗	✗	✗
Add admins	✓	✗	✗	✗
Add members	✓	✓	✗	✗
Write data	✓	✓	✓	✗
Create items	✓	✓	✓	✗
Read data	✓	✓	✓	✓

Workspace Item Permissions

- Lakehouse.
- Warehouse.
- Notebook & Spark Job Definition.
- Real-Time Intelligence.
- Data Factory.
- OneLake.

Warehouse Permissions

- Share warehouse items/data.
 - To consume warehouse data using T-SQL or Power BI.
 - "Read all data using SQL", "Read all OneLake data", "Monitor", "Audit"
 - Gets "Read" access to connect to warehouse using SQL Analytics endpoint.
- Granular Access - T-SQL GRANT statements should be used to provide object level permissions.
 - Column Level Security (CLS) – GRANT SELECT ON <table>(<col1>, <col2>) TO <user/group>;
 - Row Level Security (RLS) – Controls access to rows.
 - CREATE FUNCTION
 - CREATE SECURITY POLICY

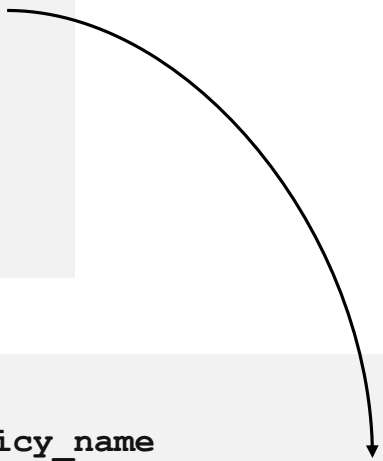
Id	Name	Salary	UID	Location
1				
2				
3				

Id	Name	Salary	UID	Location
1				
2				
3				

Warehouse Access – RLS

```
CREATE FUNCTION function_name(...)  
RETURNS TABLE  
WITH SCHEMABINDING  
AS  
<Query for function>;  
GO
```

```
CREATE SECURITY POLICY security_policy_name  
ADD FILTER | BLOCK PREDICATE function_name (column_name)  
ON table_schema_name.table_name  
WITH ( STATE = ON | OFF, SCHEMABINDING = ON | OFF );  
GO
```



Lakehouse Permissions

- Lakehouse – Share, OneLake Data Access.
- OneLake Access.
 - Table level access.
 - File/Folder level access.

Mirrored DB Permissions

- Share.
- Manage Permission.

RTI Permissions

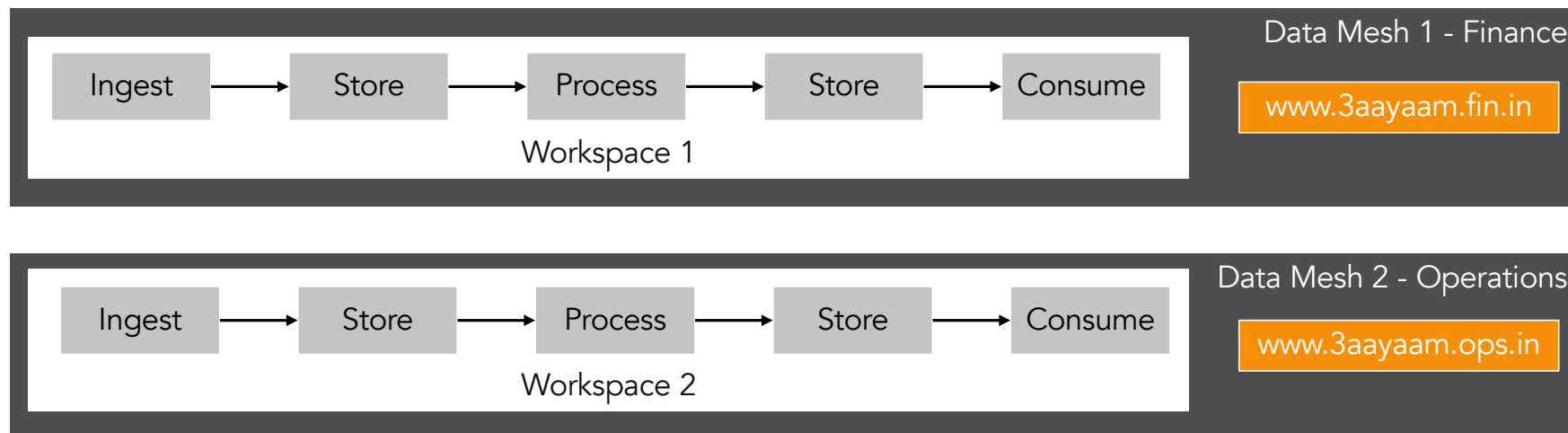
- Kusto Database → KQL Tables → Materialized Views → External Tables → Functions.
- RBAC (Role Based Access Control):
 - Admins – View, modify, remove all objects.
 - Users – View and create new objects.
 - Viewers – View objects.
 - Ingestors – Ingest data into objects. Cannot query the objects.
 - Monitors – View metadata. Schema, operations, permissions.
 - Unrestricted viewers
 - Management commands (.add, .set) can be used to grant permissions.
 - MVs, external tables and function permissions are also given using .set and .add commands.

SQL Database Permissions

- Database Roles
 - CREATE ROLE
 - ALTER ROLE
 - DROP ROLE
- GRANT, REVOKE statements.

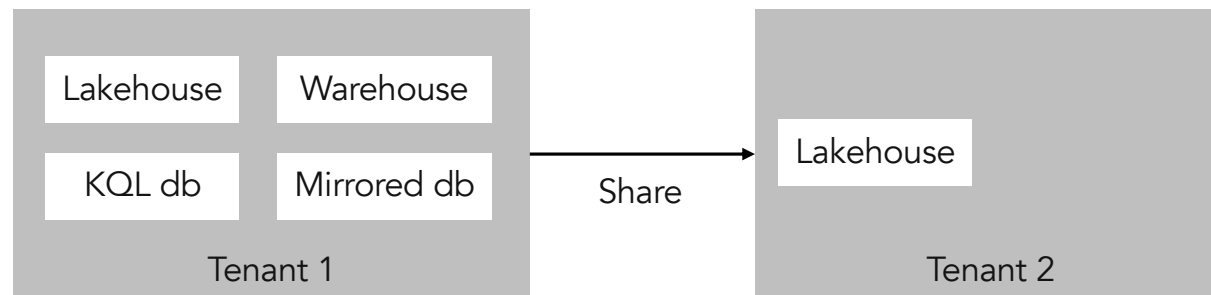
Domains in Fabric

- Revolves around Data Mesh architecture. Decentralized data architecture that organizes data by specific business domains (Marketing, Finance, IT, Operations etc.).
 - Each business domain would have their own data engineering pipeline.
 - Federated governance. Tenant level governance can be delegated to domain level.



Domains in Fabric

- Multiple Workspaces can be associated with a single domain.
- Subdomains.
- Domain Roles:
 - Fabric Admin – Create, edit, add users, associate workspaces etc.
 - Domain Admin – Can access domain tab in admin portal. Add workspaces in domain.
 - Domain Contributors – Workspace admins via domains. Assign workspaces to domain.
- External Data Sharing – Share data from one fabric tenant to another fabric tenant.



Lineage

- Shows relationship between all items in a workspace.
- Where data has come from, where is it moving etc.

Capacity Access

- Capacity Admins
 - Assign workspaces to the capacity
 - Manage user permission to the capacity
 - Manage workloads to configure memory usage